# 9.2.5 Diagonalizing a matrix

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Week 09 due Mar 19, 2025 07:24 IST

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x_1, and so forth, so here we get lambda_n-1 times x_n-1. And on the

Left we know that multiplying a matrix times the individual columns just means

multiplying that matrix times individual columns. And if we now set columns on the

left equal to columns on the right, what we get is exactly the A times x_0 is

equal to lambda_0 times x_0 and so forth. So from this, we conclude that if we have

such a matrix X then

3:27 / 3:27

2.0x

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## Reading assignment

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Read Unit 9.2.5 of the notes. [LINK]

- [x] done

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vector v tilde. So what does that really mean? All we've done here is we've

replaced v with X times v tilde, where we now in v tilde have the coefficients in

terms of this basis that we get from the columns of our matrix X. And we can

obviously do the exact same thing to w. We can say, "Let's also view that vector

in the same basis." And then this here becomes the vector w tilde. So what do we

have now? We have

5:01 / 5:01

2.0x

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## Homework 9.2.5.1

1/1 point (graded)

In Homework 9.2.3.3, we computed the eigenvalues and corresponding eigenvectors of

$$A = \begin{pmatrix} -2 & 3 & -7 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Use the answer to that question to give a matrix $X$ such that $X^{-1}AX = \Lambda$. Check that $AX = X\Lambda$.

- [x] done

**Solution:**

For a complete solution see the notes.

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Answers are displayed within the problem

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raising this to the kth power. And if you think about it, if you multiply these multiply these

together the X inverse cancels the X every time. And therefore this is

actually the same as A times Lambda to the kth power times X inverse times v. Now

raising a diagonal matrix to the kth power is easy. That's a matter of raising each

of the individual diagonal elements to the kth power. So now what you can do is if

you want to apply A

2:10 / 2:10

2.0x

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## Homework 9.2.5.2

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Let $A \in \mathbb{C}^{m \times m}$ have distinct eigenvalues.

ALWAYS/SOMETIMES/NEVER: $A$ is diagonalizable.

ALWAYS ✔ Answer: ALWAYS

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